Transferring pretrained audio models to artificial noise and abiotic sound

Experiment brief

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OVERVIEW

Do frozen pretrained audio models transfer to noise in ARU data?

- **Classes:** 14 target classes in ARU recordings, 11 artificial and 3 abiotic; 9 animal and other classes serve only as negatives
- **Models:** BirdNET 2.4, BirdNET 3.0, Perch v8 and Perch 2.0 (bird); CNN14 (AudioSet); ResNet-18 (ImageNet), all frozen
- **Adaptation:** a small MLP head on the embeddings, trained on N = 5–160 clips per class, with 10 random training draws per N
- **Test:** 50 clips per class from recordings never used for training
- **Zero-shot:** the model's own label for the class, where one exists

TABLE 1 · DATA

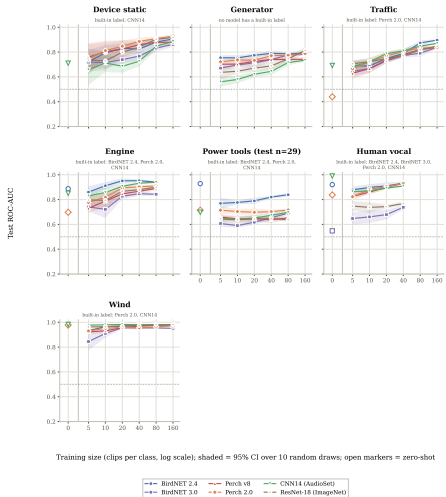
Fourteen noise classes, split by source recording

Class	Type	Recordings / clips	Test rec. / clips	Training rec. / clips	Max N
Device static	artificial	168 / 1,416	9 / 50	159 / 1,366	160
Generator	artificial	32 / 437	10 / 50	22 / 384	160
Traffic	artificial	53 / 297	13 / 50	40 / 242	160
Engine	artificial	115 / 170	38 / 50	76 / 117	80
Power tools	artificial	11 / 162	9 / 29	2 / 133	80
Human vocal	artificial	16 / 117	14 / 50	2 / 64	40
Metal clanging *	artificial	28 / 60	13 / 30	15 / 30	20
Airplane *	artificial	12 / 52	6 / 25	6 / 27	20
Gun *	artificial	7 / 19	5 / 9	2 / 10	10
Fireworks *	artificial	4 / 13	2 / 6	2 / 7	5
Human non-vocal *	artificial	2 / 8	1 / 3	1 / 5	5
Wind	abiotic	106 / 508	11 / 50	94 / 453	160
Thunder *	abiotic	5 / 21	2 / 8	3 / 13	10
Water *	abiotic	3 / 11	2 / 5	1 / 6	5

* Low-sample class (fewer than 90 clips; provisional). Power tools and human vocal have only 2 training recordings each.

FIGURE 1 · HOW MANY CLIPS?

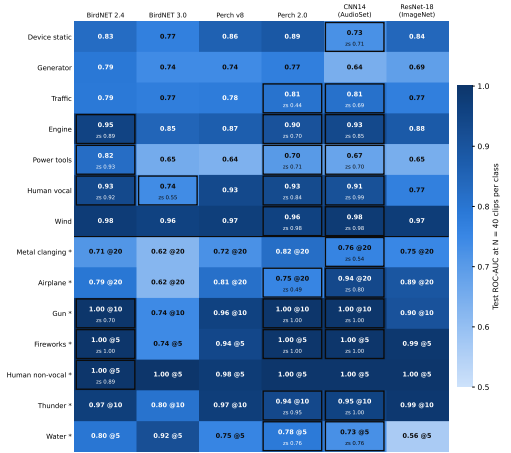
Performance levels off near N = 40 for most classes



- Most curves come within 0.02 AUC of their best by N = 40
- Device static and traffic are still rising at N = 160
- Going from 40 to 80 clips adds 0.01–0.04 AUC
- Open markers at N = 0: the model's own label (zero-shot)

FIGURE 2 · SUMMARY AT N = 40

The class matters more than the model

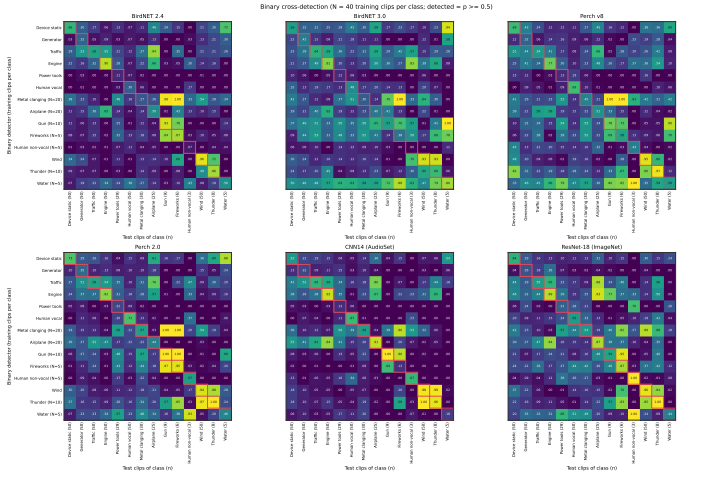


Outlined = model has a built-in label for the class (zs = its zero-shot AUC); * low sample (provisional), @N = largest N reached.

- Class means range from 0.97 (wind) to 0.69 (power tools)
- Model means range from 0.78 (BirdNET 3.0) to 0.87 (BirdNET 2.4)
- The trained head beats the built-in label in 13 of 29 known pairs (median N = 5)
- No model has a label for generator

FIGURE 3 · CROSS-DETECTION

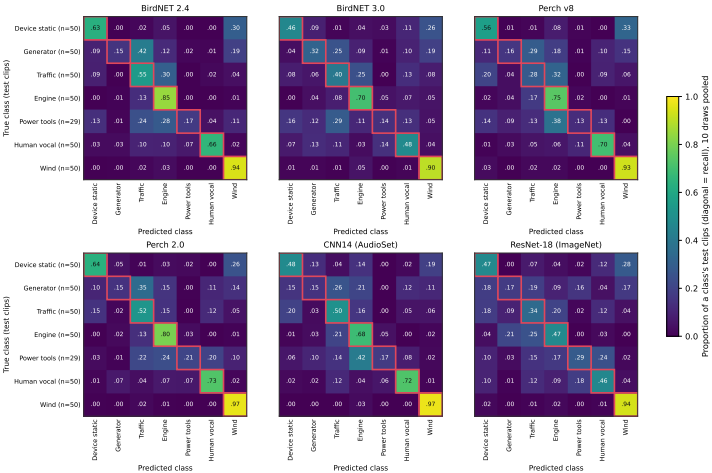
Binary detectors respond to similar noise



- Detected = $p \geq 0.5$; diagonal = recall
- Mean recall 0.55–0.62; false alarms 0.14–0.23
- Groups: impulsive sounds, wind and thunder, vehicles

FIGURE 4 · MULTICLASS

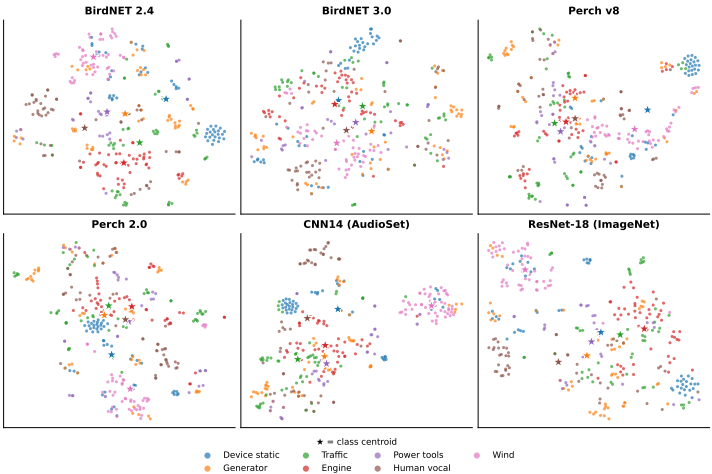
One head recognises wind, not generator or power tools



- 7 classes with ≥ 40 training clips
- Wind recall 0.90–0.97
- Generator and power tools 0.13–0.32, mostly called traffic or engine
- Device static is often called wind
- Macro F1 0.43–0.55

FIGURE 5 · EMBEDDINGS

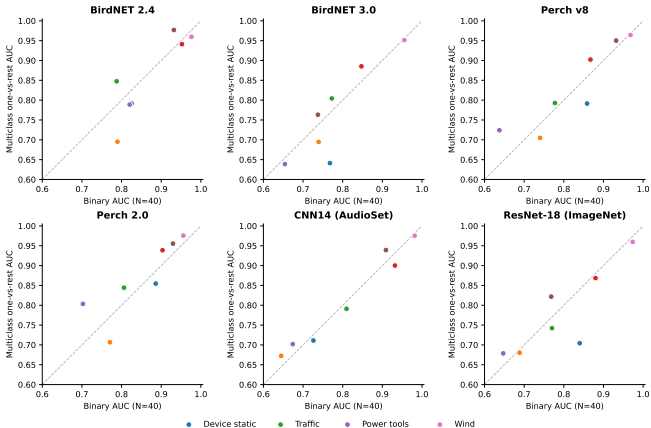
The confusions are already in the embeddings



- t-SNE of raw test-clip embeddings; no classifier
- Device static is compact in most models
- Generator, traffic, engine and power tools overlap

FIGURE 6 · BINARY VS MULTICLASS

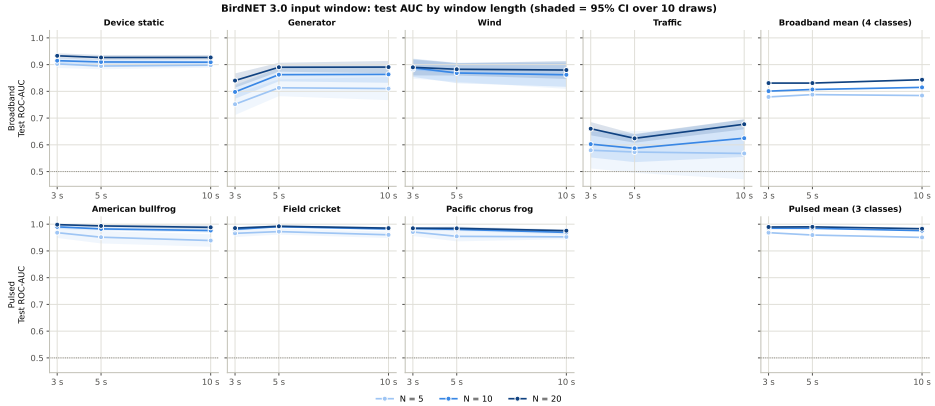
Both discriminate each class about equally well



- Same 40 training clips and the same positive test clips
- Mean difference -0.002 (range -0.14 to $+0.10$)
- Largest drop: device static, for which wind is a hard negative

FIGURE 7 · BIRDNET 3.0 WINDOW LENGTH

Longer input windows help only generator



At N = 20, generator rises from 0.84 (3 s) to 0.89 (5 and 10 s). Device static and wind are unchanged, traffic shows no trend, and pulsed sounds stay at 0.98–0.99.

SUMMARY

Take-home

1. About 40 labelled clips per class brings most classes close to their best performance
2. The class matters more than the model; more clips from the same few recordings don’t help
3. A trained head usually matches or beats a model’s built-in label
4. Errors fall between acoustically similar sounds, in binary detectors, the multiclass head and the embeddings alike
5. Longer input windows help only generator

Supplementary material

TABLE S1

Pretrained models

Model	Embedding	Input	Built-in labels used for zero-shot
BirdNET 2.4	1024-D	3 s at 48 kHz	Engine, Power tools, Human vocal, Human non-vocal, Gun, Fireworks
BirdNET 3.0	1280-D	3 s at 32 kHz (3, 5 and 10 s in Figure 7)	Human vocal
Perch v8	1280-D	5 s at 32 kHz (padded)	none
Perch 2.0	1536-D	5 s at 32 kHz (padded)	FSD50K classes (Table S2)
CNN14 (AudioSet)	2048-D	3 s at 32 kHz	AudioSet classes (Table S2)
ResNet-18 (ImageNet)	512-D	mel spectrogram, 3 s at 16 kHz	none

TABLE S2

Built-in labels used for zero-shot (direct matches only)

Class	BirdNET 2.4	BirdNET 3.0	Perch 2.0	CNN14
Device static				Static
Engine	Engine		Engine; Idling	Engine; Idling
Traffic			Traffic noise and roadway noise	Traffic noise, roadway noise
Power tools	Power tools		Power tool; Drill	Power tool; Drill
Human vocal	Human vocal	Human vocal	Speech; Human voice; Homo sapiens	Speech
Metal clanging				Clang
Airplane			Fixed-wing aircraft and airplane	Fixed-wing aircraft, airplane
Gun	Gun		Gunshot and gunfire	Gunshot, gunfire
Fireworks	Fireworks		Fireworks	Fireworks
Human non-vocal	Human non-vocal			
Wind			Wind	Wind; Wind noise (microphone)
Thunder			Thunder	Thunder
Water			Water	Water

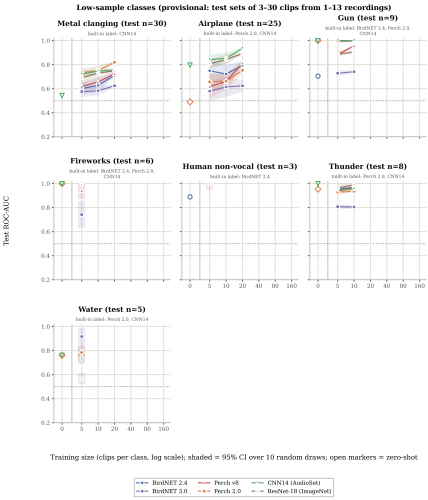
TABLE S3

Clips available for the window-length experiment

Class	Group	Clips (recordings)
Device static	broadband	239 (147)
Generator	broadband	79 (28)
Wind	broadband	79 (79)
Traffic	broadband	48 (36)
American bullfrog	pulsed	378 (13)
Field cricket	pulsed	539 (103)
Pacific chorus frog	pulsed	705 (41)

FIGURE S1 · LOW-SAMPLE CLASSES

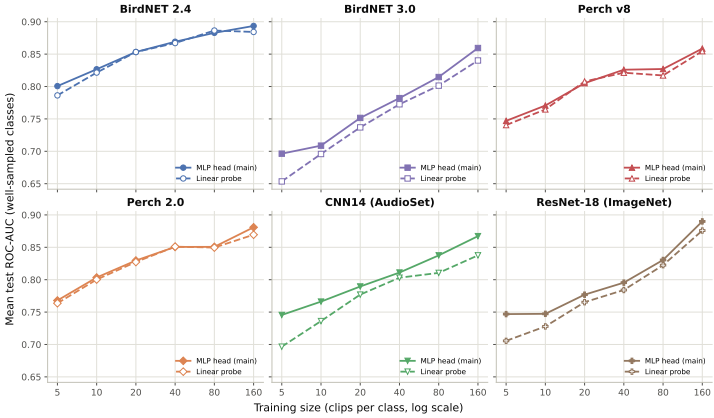
Saturation curves for the low-sample classes (provisional)



- Test sets of 3–30 clips from 1–13 recordings
- Curves end at N = 5–20
- The CIs cover training draws only, so they understate test-set uncertainty

FIGURE S2 · HEAD CHOICE

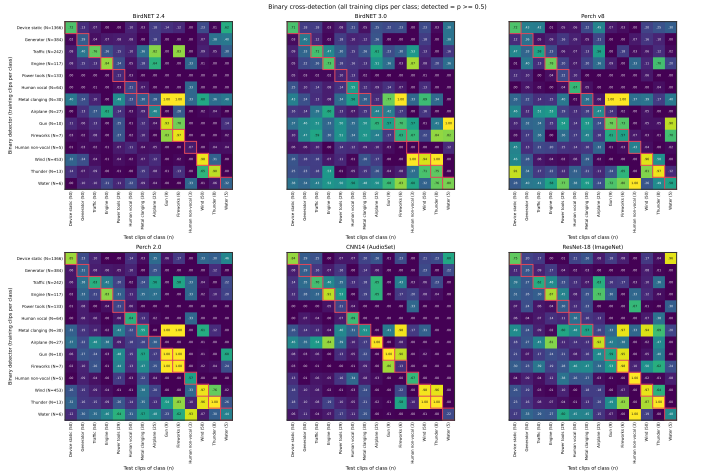
A linear probe gives the same curves as the MLP head



- The MLP is ahead by 0.003–0.026 AUC; ranking unchanged
- The linear probe is fitted to convergence at every N and shows the same rise with N

FIGURE S3 · CROSS-DETECTION, ALL DATA

Training on all clips mainly reduces false alarms



- False alarms fall from 0.14–0.23 to 0.11–0.17
- Recall rises for device static and traffic
- No gain for power tools or human vocal (2 recordings each)